

SCSD2613

System Analysis and Design



PART IV

The Analysis Process #3: Structure Chart & Entity Relationship Diagram

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■ OBJECTIVES

- 1. Understand the purpose of structure chart.**
- 2. Able to transform DFD to structure chart.**
- 3. Understand the basic ERD.**

■ MAJOR TOPICS

STRUCTURE CHART

- Structure chart symbols
- Guideline for drawing structure chart
- Translating DFD to Structure Chart

ENTITY RELATIONSHIPS DIAGRAM (ERD)

- ERD basic concepts
- DFD and ERD

PART 1

- Structure Chart

■ STRUCTURE CHART

- A top-down representation of business functions and processes.
- Also called a functional decomposition diagram.
- Show the program modules and the relationships among them.

■ STRUCTURE CHART SYMBOLS



Modules



Library modules



Module call



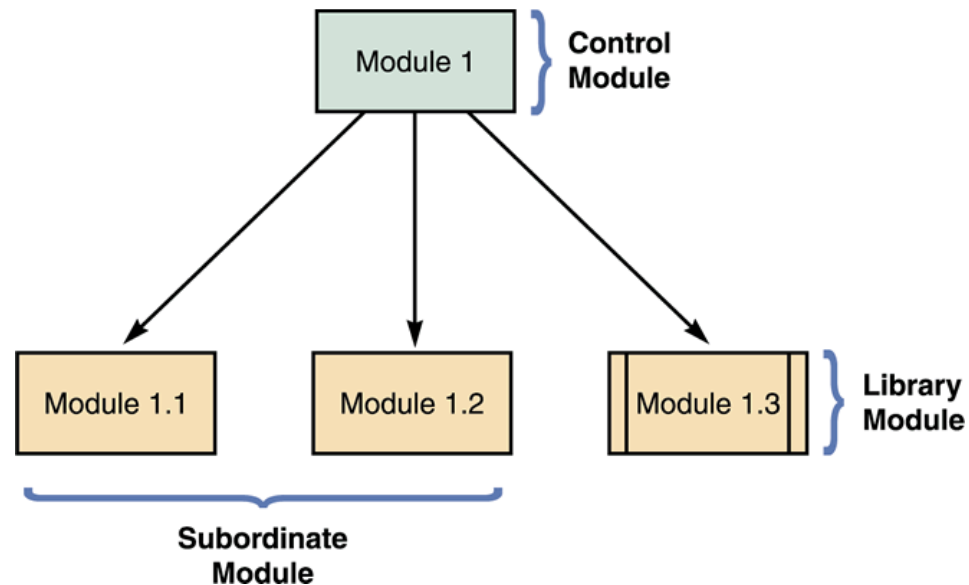
Data couple



Flag

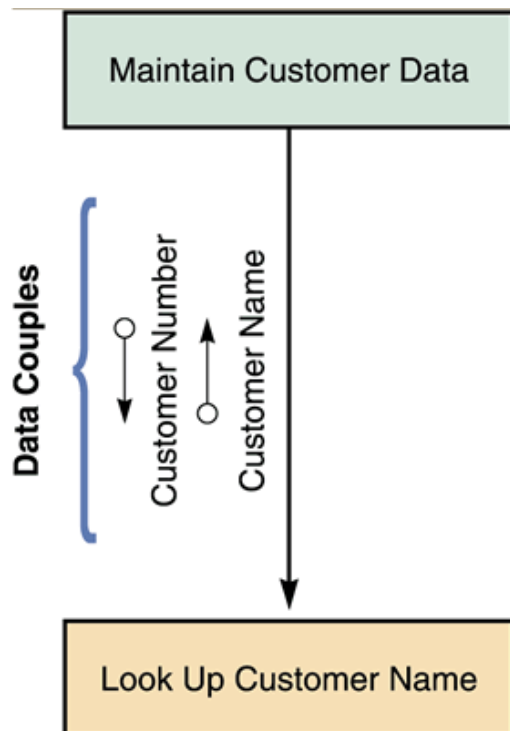
■ STRUCTURE CHART SYMBOLS

- Module type:
 - **CONTROL MODULE** - a higher-level module that directs lower-level modules
 - **SUBORDINATE MODULES** – a lower-level module in a structure chart.
 - **LIBRARY MODULE** - a module that is reusable and can be invoked from more than one point in the chart.

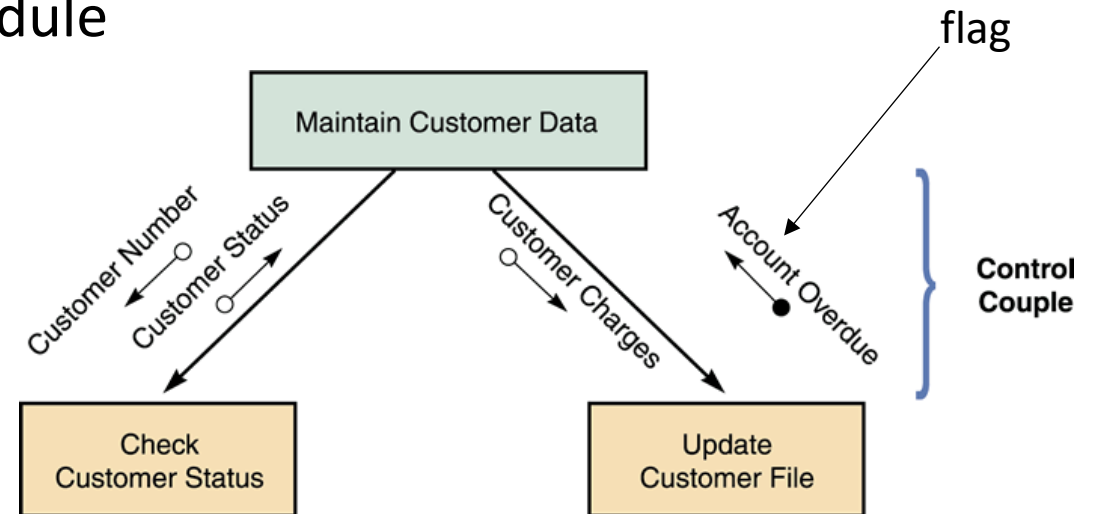


■ STRUCTURE CHART SYMBOLS

- **DATA COUPLE** (arrow with an empty circle) - shows data that one module passes to another.



- **CONTROL COUPLE**- shows a message (flag) which one module sends to another.
- **FLAG** - is a message sent from one module to another. A module uses a flag to signal a specific condition or action to another module



■ STRUCTURE CHART SYMBOLS

• CONDITION

- A specified action or state in a structure chart.
- A condition line indicates that a control module determines which subordinate modules will be invoked, depending on a specific condition

An example of structure chart condition

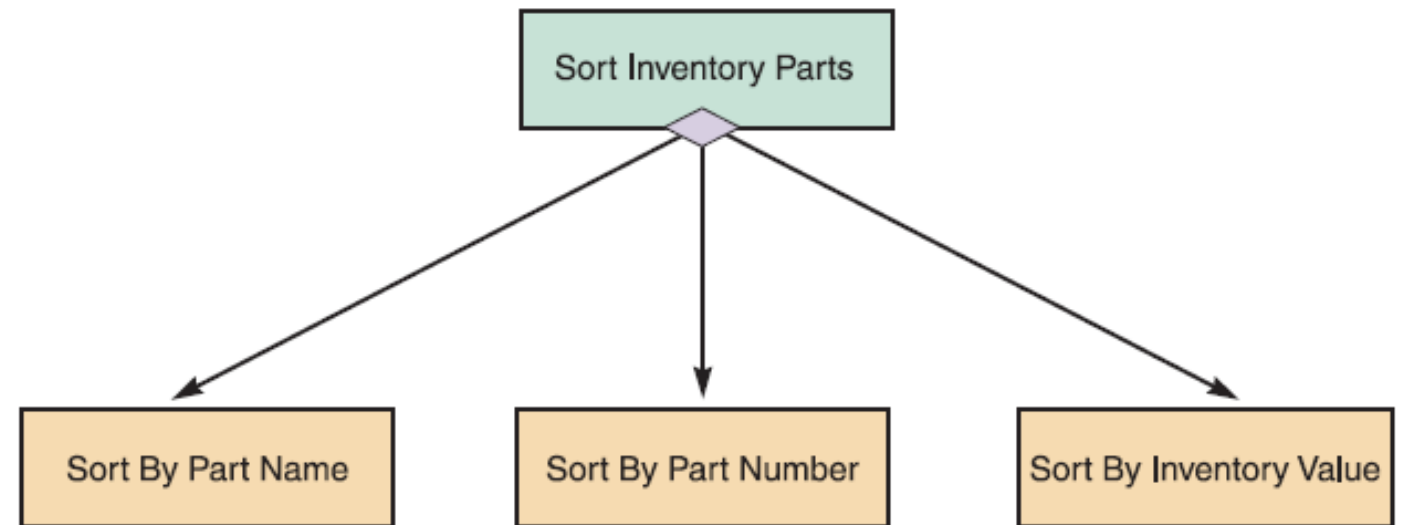


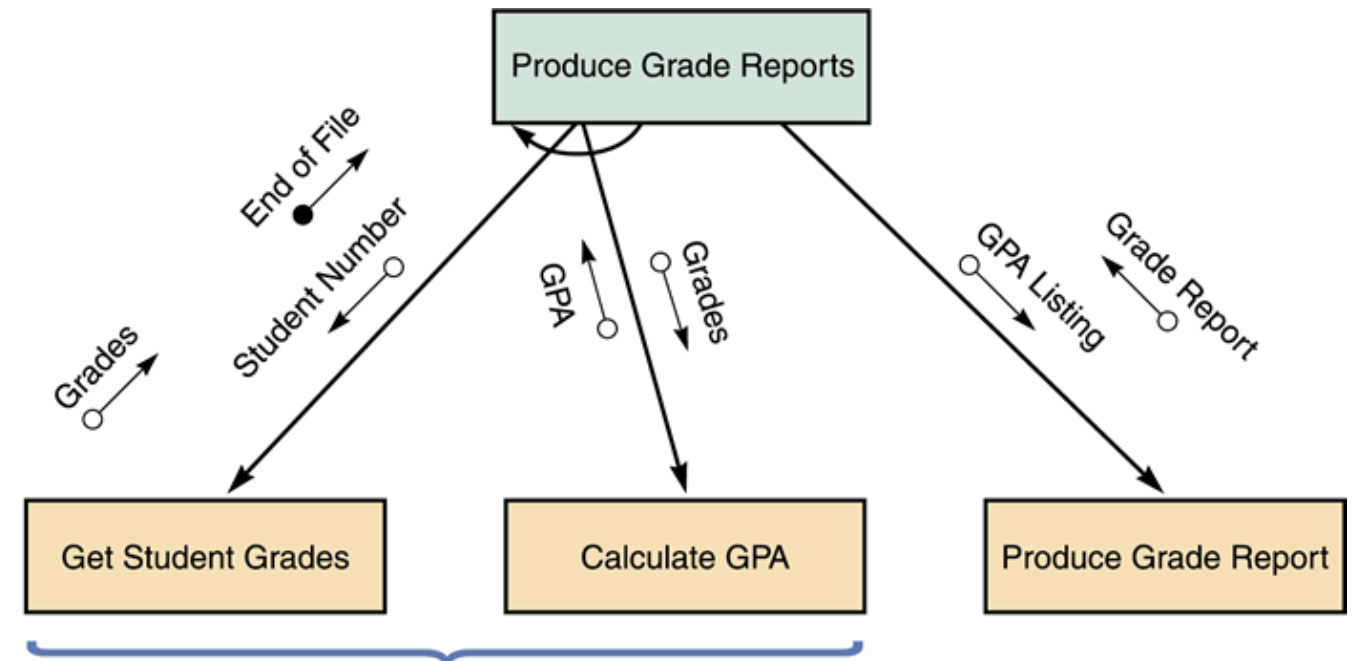
FIGURE 11-13 The diagram shows a control module that triggers three subordinate modules.

■ STRUCTURE CHART SYMBOLS

• LOOP

- indicates that one or more modules are repeated

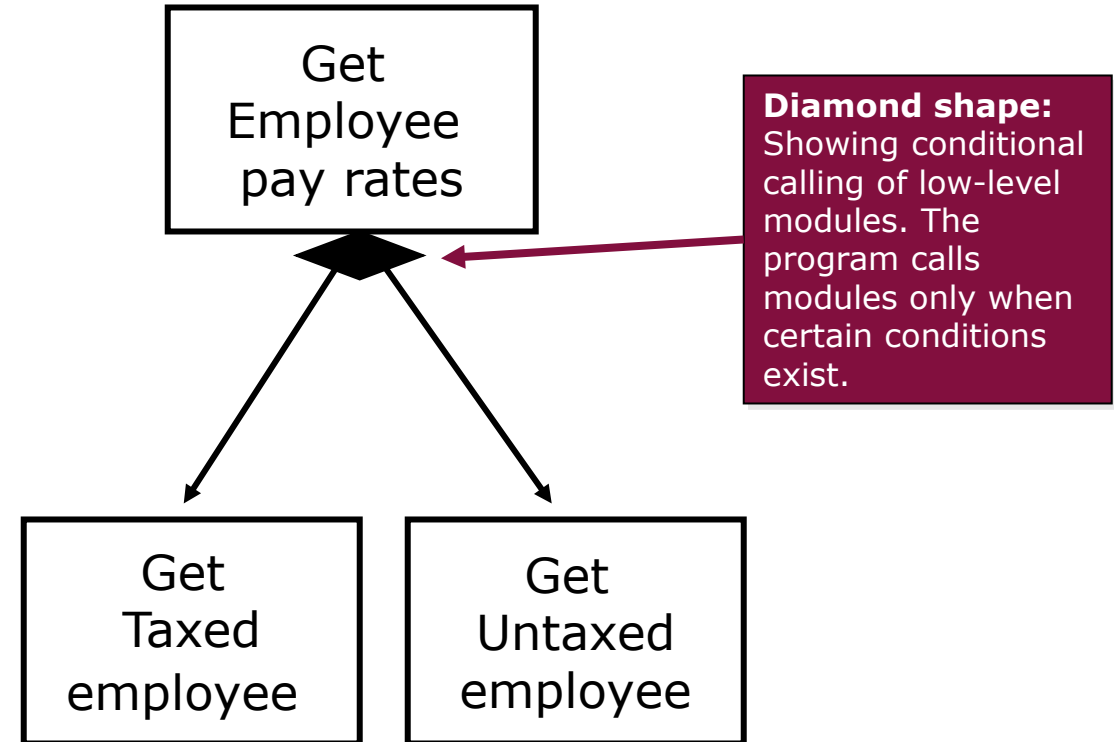
An example of structure chart loop



The curved arrow indicates that these modules are repeated.

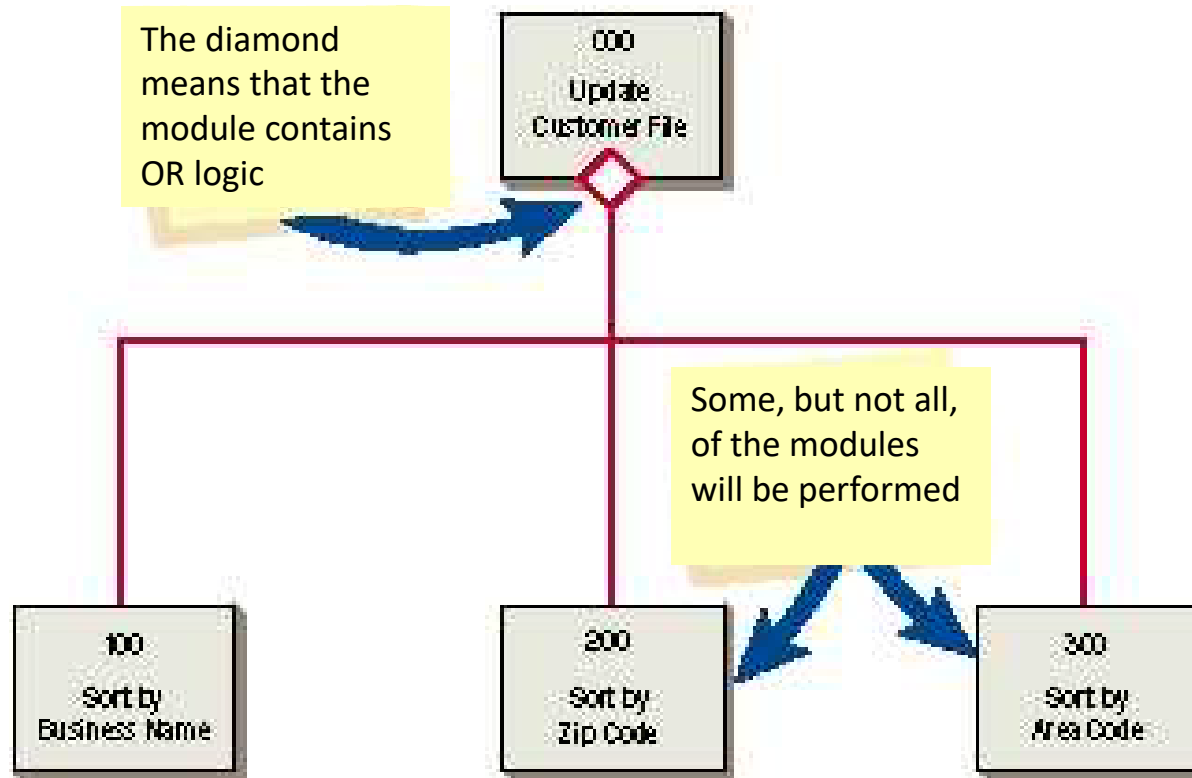
■ STRUCTURE CHART – OR Logic

```
START Module Get Employee pay rates
If employee >= 3000
    ...
Else if employee <3000
    ...
END if
END START
```



STRUCTURE CHART – OR Logic

Example : Prepare Customer Bill

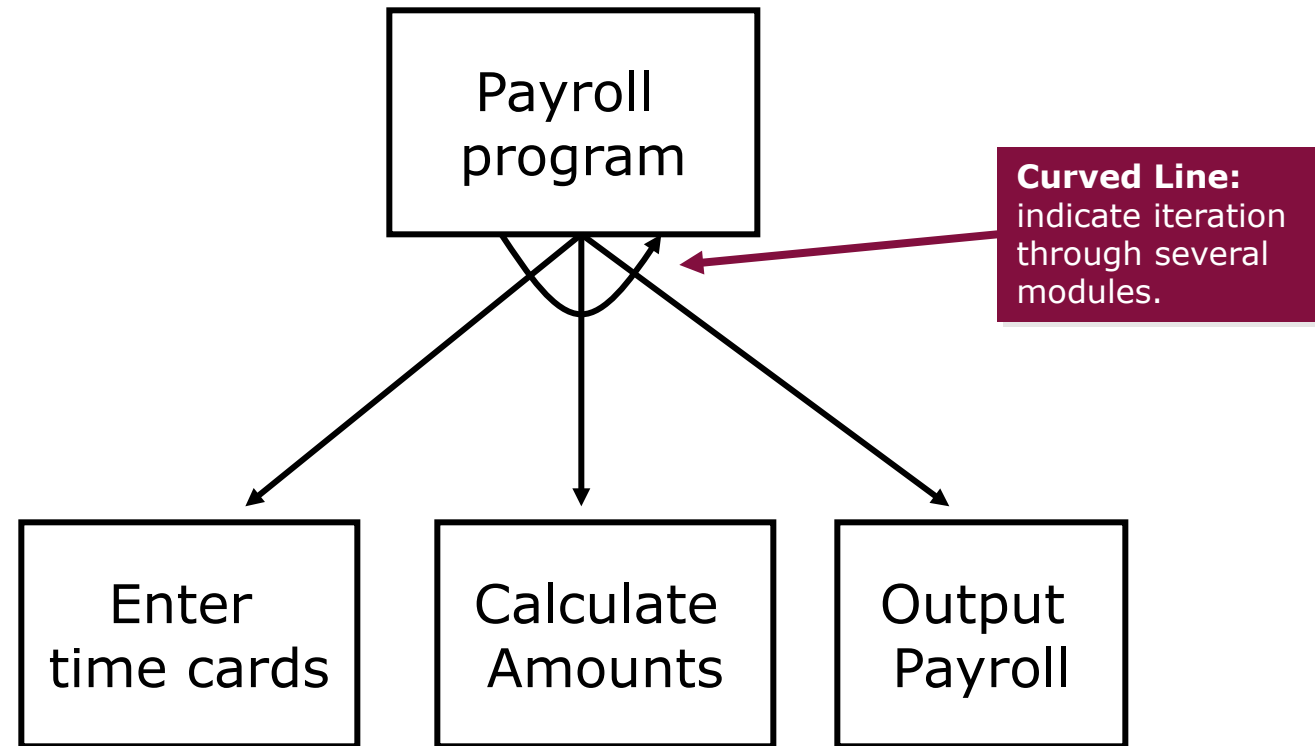


```

Start Module 000
If  update = BusName
    Do module 100
Elseif  update = ZipCode
    Do module 200
Else
    Do module 300
End Module A
  
```

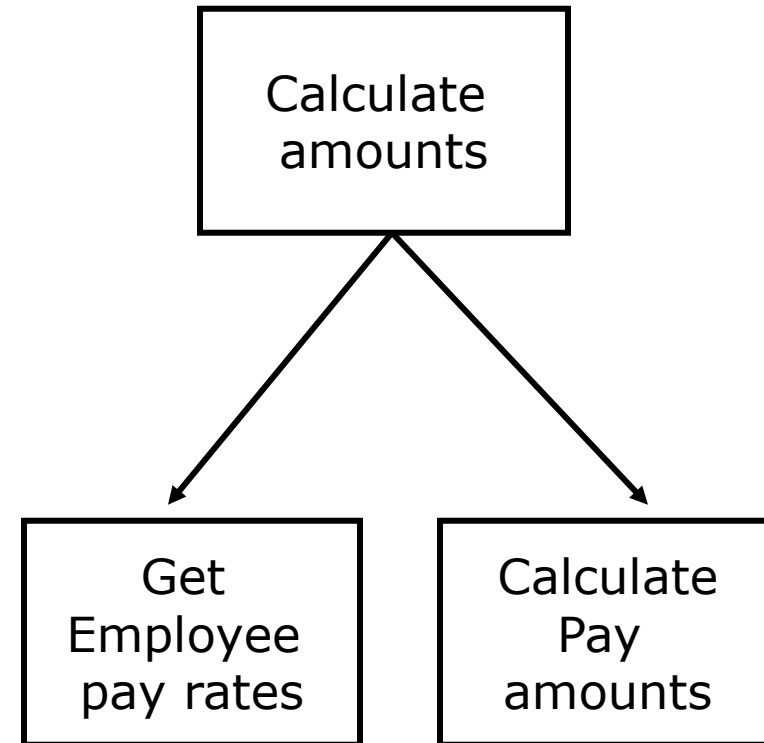
■ STRUCTURE CHART – WHILE Logic

```
START Program
DoUntil No more time cards
    Call Enter time cards
    Call Calculate amounts
    Call Output payroll
END Until
END START
```



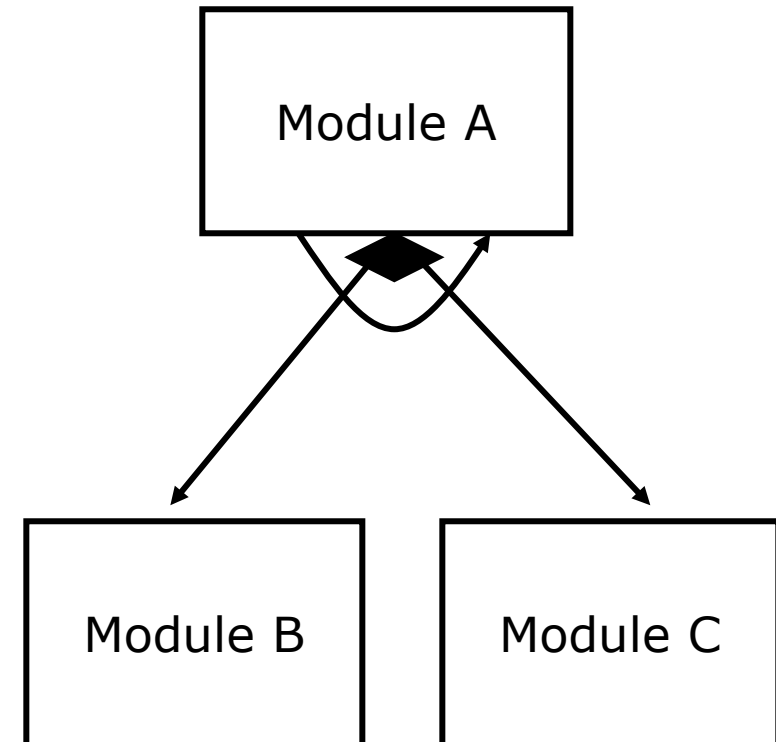
■ STRUCTURE CHART – without OR & WHILE Logic

```
START Module Calculate amounts  
  Call Get employee pay rates  
  Call Calculate pay amounts  
END START
```



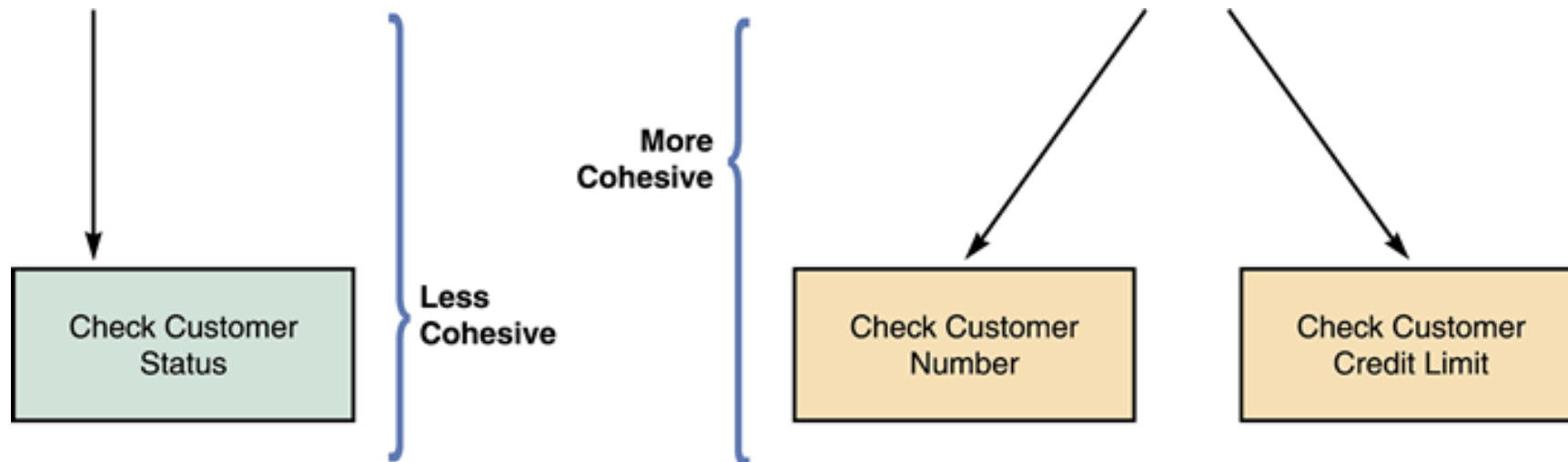
■ STRUCTURE CHART – with OR & WHILE Logic

```
START Module A  
Repeat  
If...  
  Do Module B  
Else...  
  Do Module C  
END Repeat  
END START
```



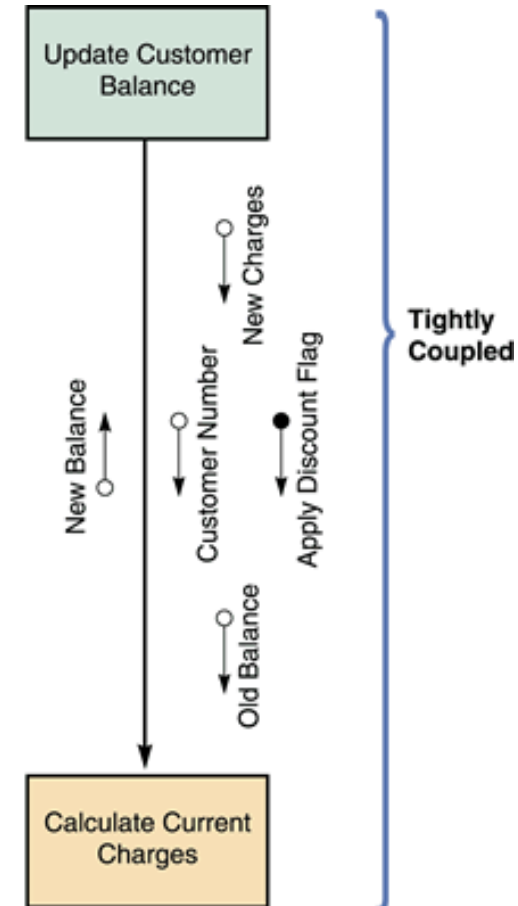
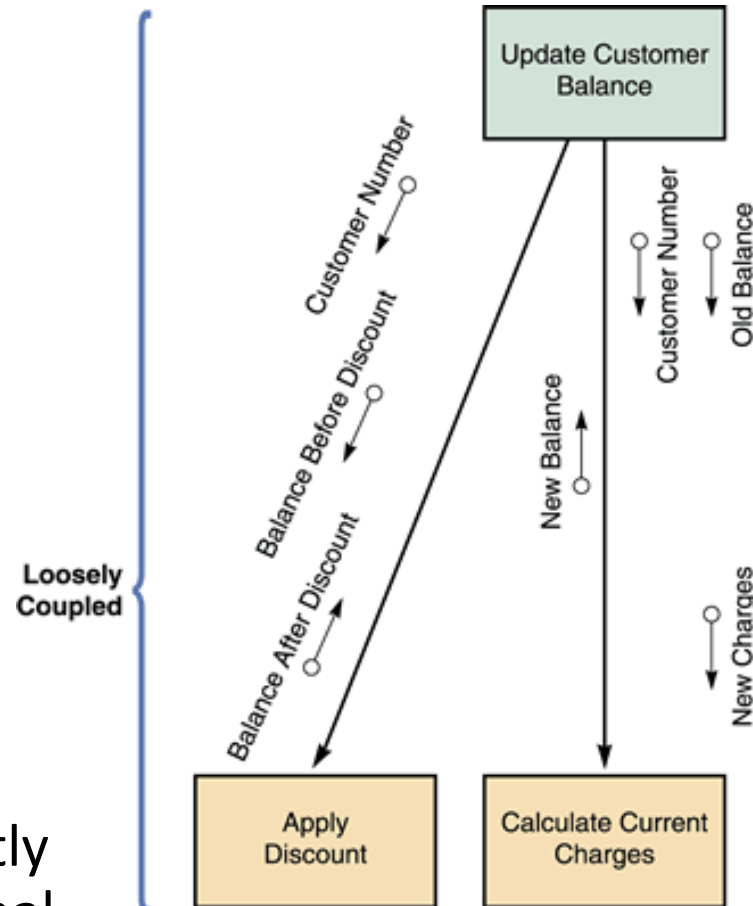
■ STRUCTURE CHART - COHESION

- Measures a module's scope and processing characteristics.
- A module that performs a single function or task has a high degree of cohesion, which is desirable.

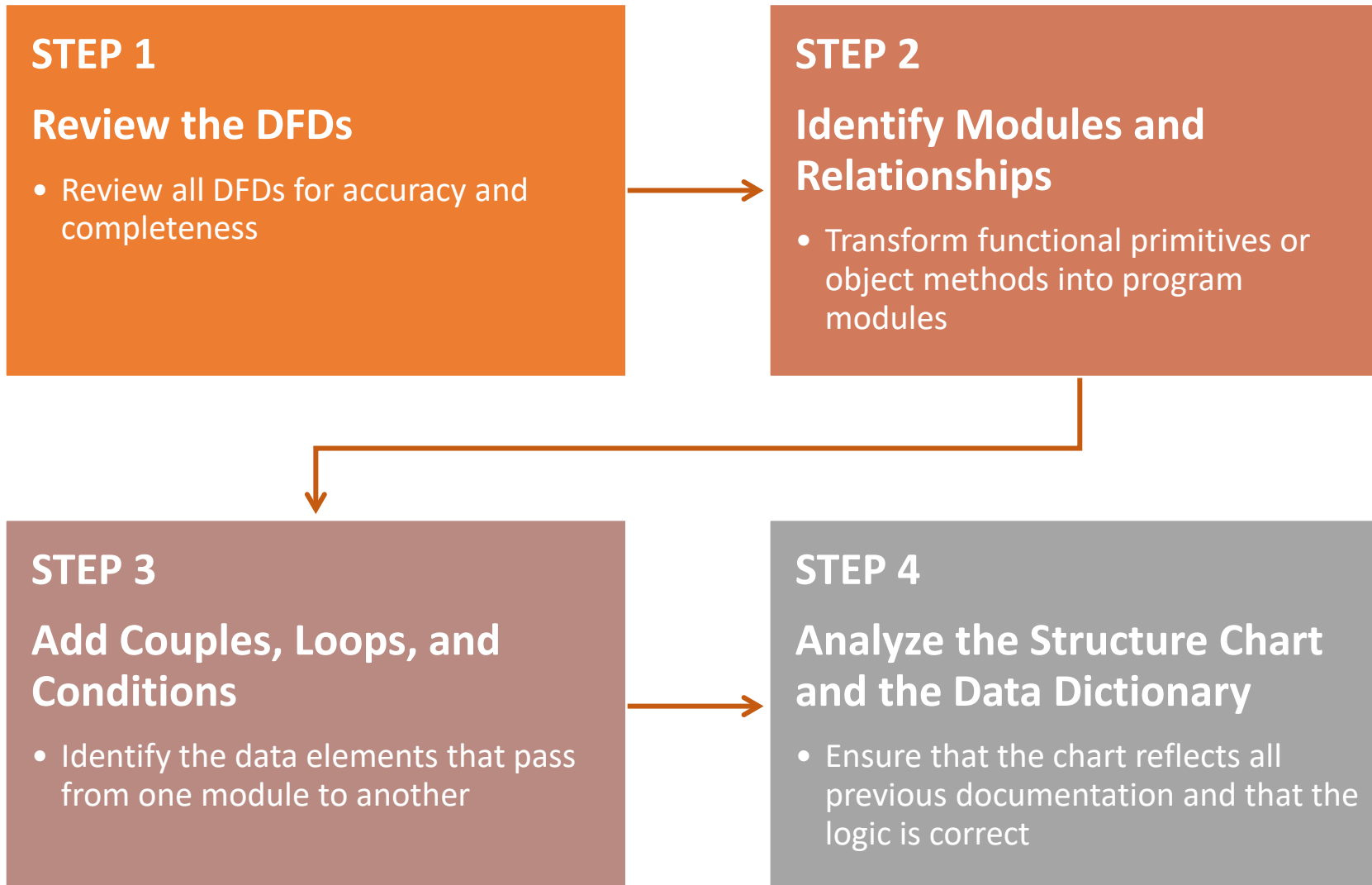


■ STRUCTURE CHART - COUPLING

- A specified action or state in a structure chart.
- If you need to make a module more cohesive, you can split it into separate units, each of which performs a single function
 - Loosely coupled : Modules that are relatively independent. Loosely coupled modules are easier to maintain and modify, because the logic in one module does not affect other modules.
 - Tightly coupled : If modules are tightly coupled, one module refers to internal logic contained in another module.



■ GUIDELINE FOR DRAWING A STRUCTURE CHART



■ TRANSLATING DFD TO STUCTURE CHART

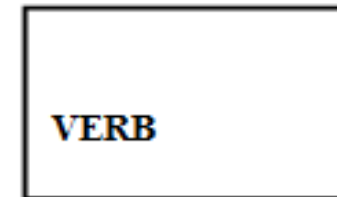
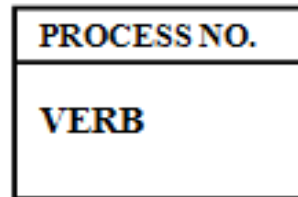
<u>ELEMENTS</u>	<u>DFD</u>	<u>S/CHART</u>
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ENTITIES

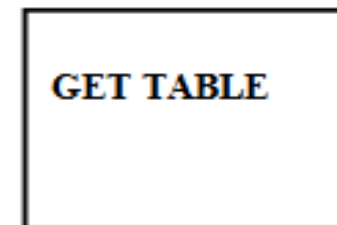
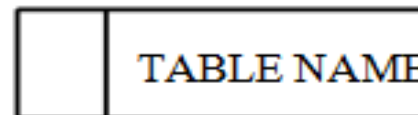


NONE

PROCESS



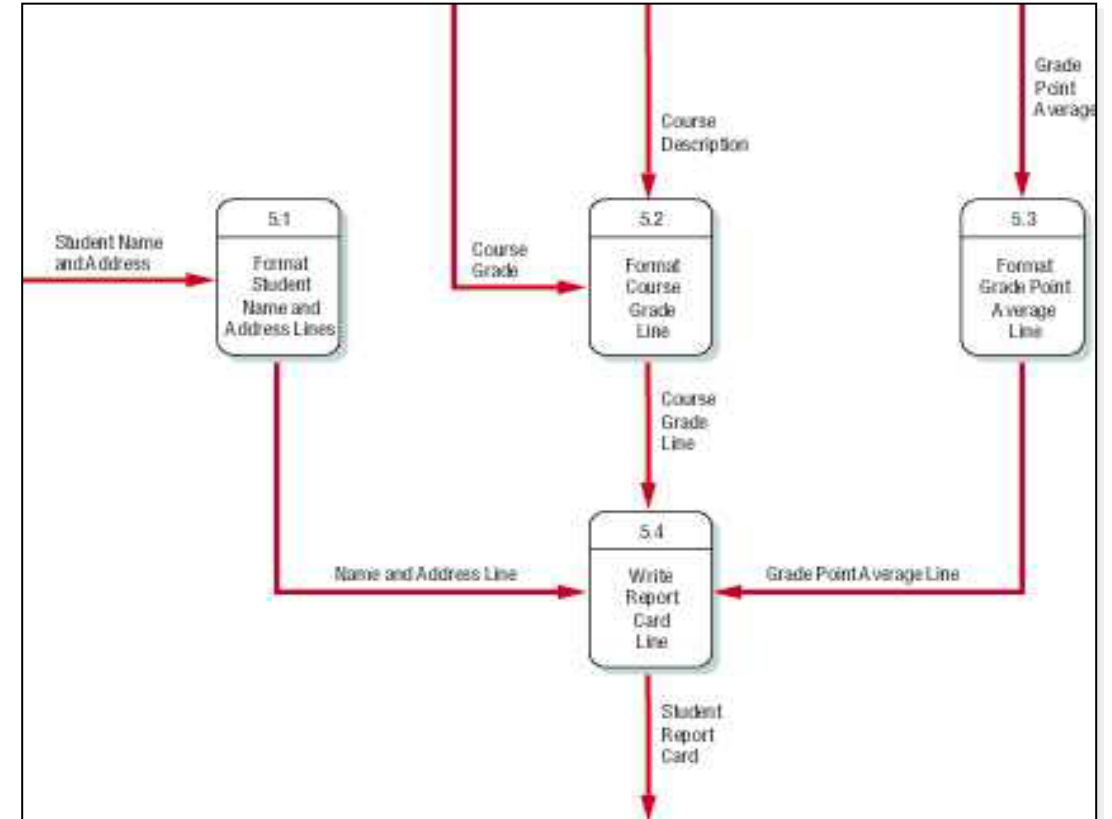
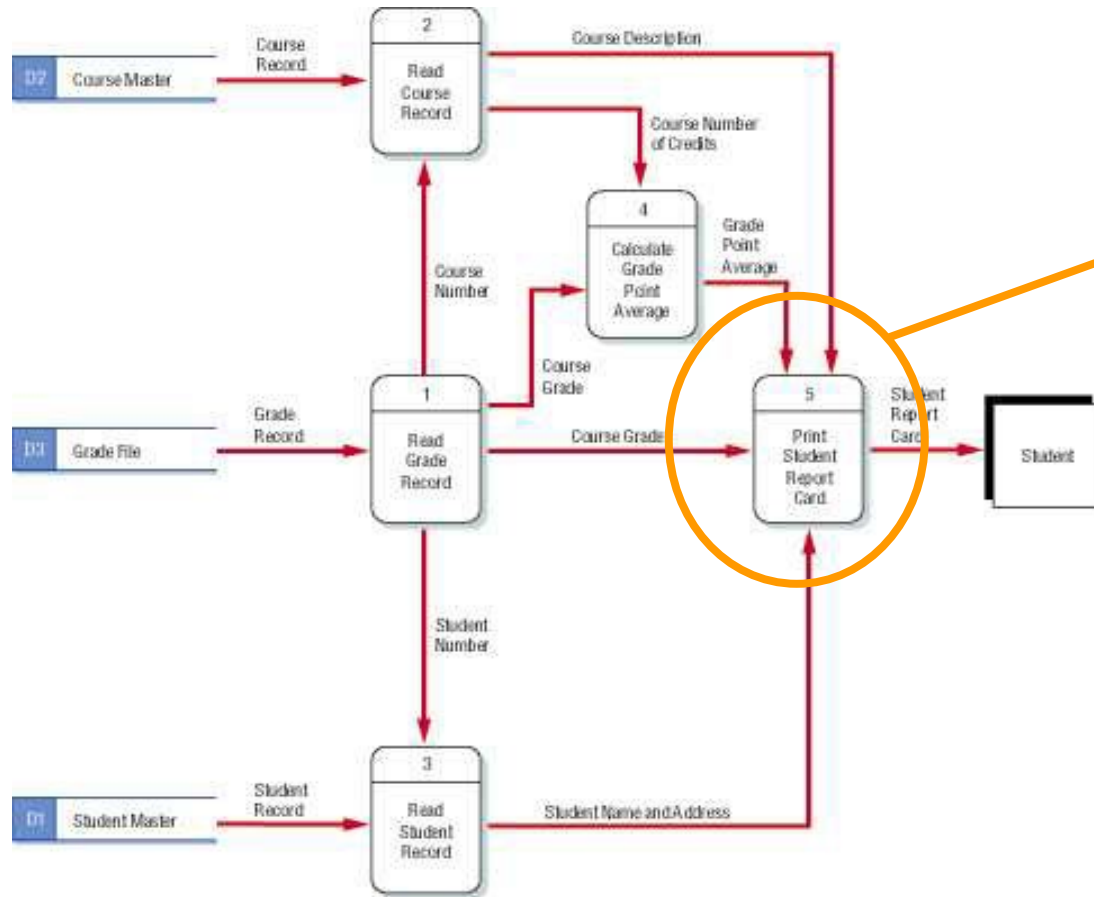
FILE



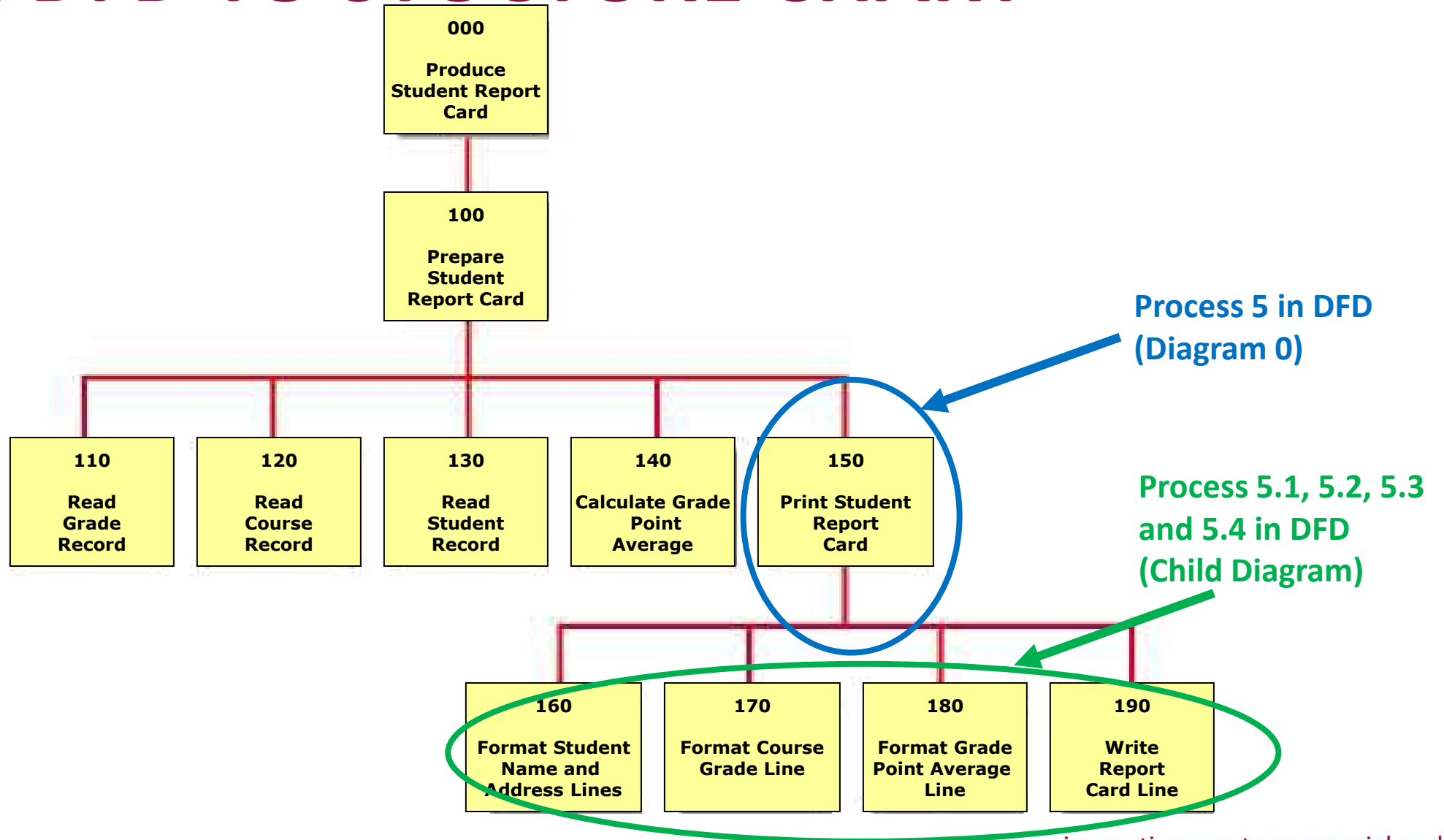
■ TRANSLATING DFD TO STUCTURE CHART

<u>ELEMENTS</u>	<u>DFD</u>	<u>S/CHART</u>
DATA FLOW	 DATA FLOW NAME	 DATA FLOWNAME
FLAG	NONE	 FLAG NAME/LABEL
CALL	NONE	

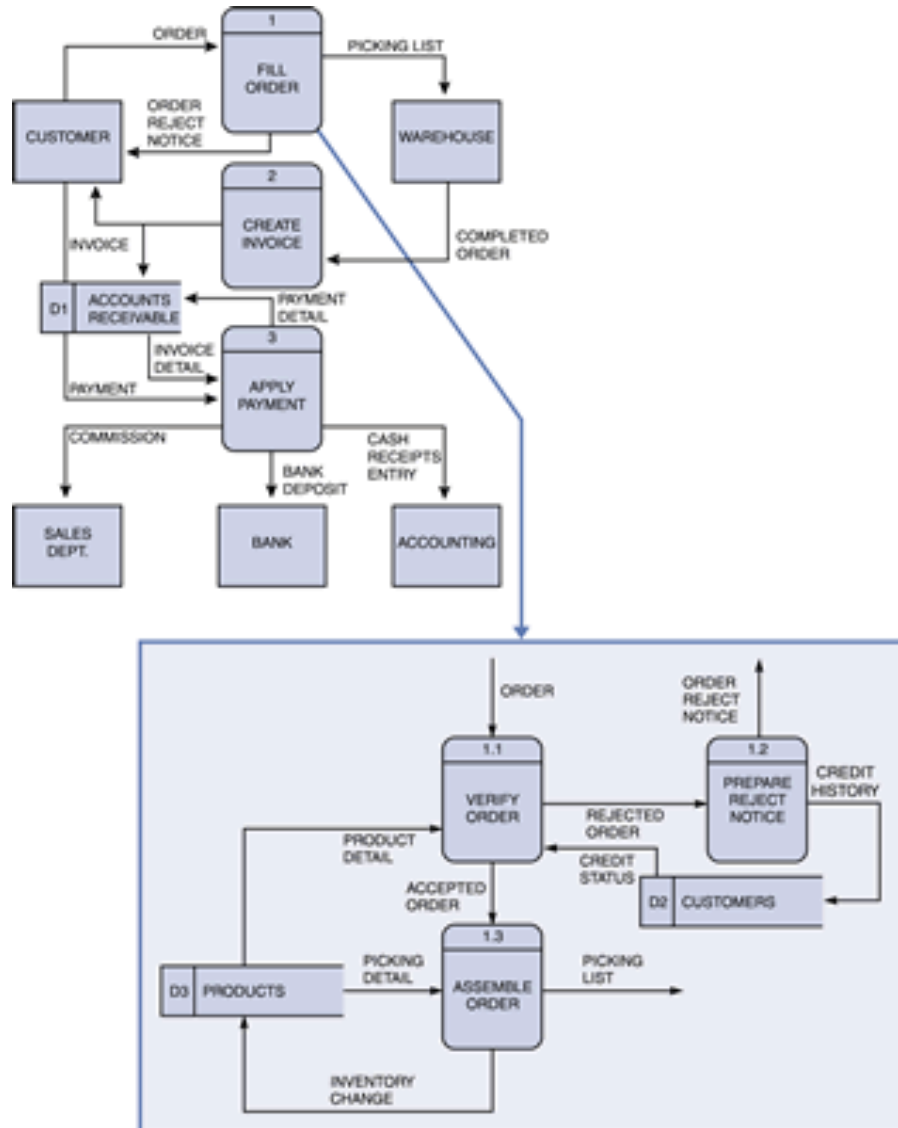
FROM DFD TO STUCTURE CHART



FROM DFD TO STUCTIONE CHART



DFD TO STUCTURE CHART Example





LET'S TRY

Checkpoint 1: DFD to Structure Chart (Final 2016/2017)

Figure 1 shows the physical Data Flow Diagram for Book Rental System. Based on Figure 1, convert the DFD into a structure chart.

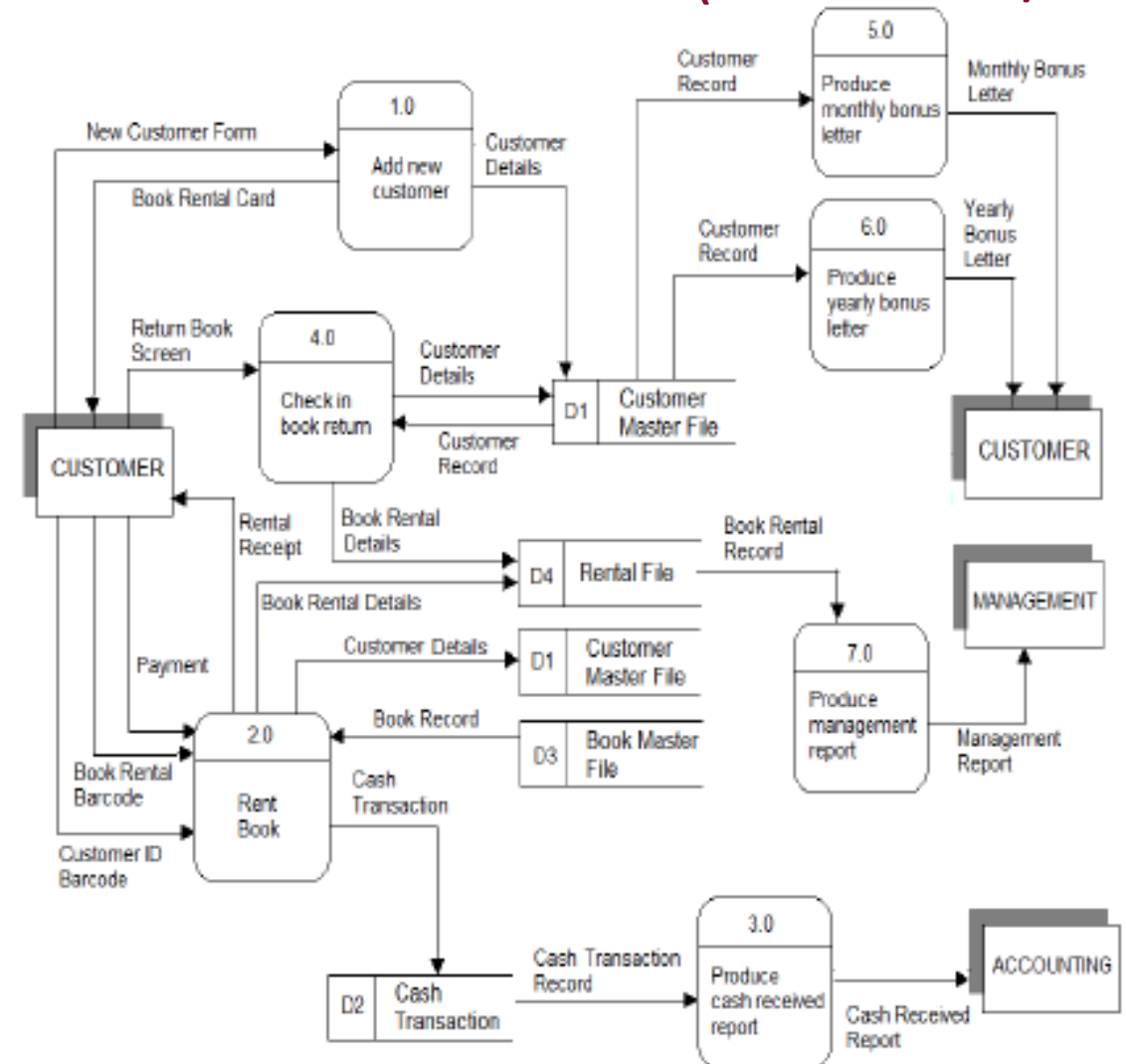


Figure 1: Physical DFD for Book Rental System

PART 2

- Entity Relationship Diagram

■ WHAT IS ER DIAGRAM?

- A diagram used to define the data storage requirements is called the entity-relationship diagram (ERD)
- Data storage requirements include the data entities, their attributes, and the relationships among the data entities.

■ ERD BASIC CONCEPTS

CONCEPTS	DESCRIPTIONS
Entity type	Group of objects with same properties, identified by enterprise as having an independent existence. A class of persons, places, objects, events, or concepts about which we need to capture and store data. Examples are person, object, place, event, and concepts.
Entity occurrence	Uniquely identifiable object of an entity type.
Relationship type	Set of meaningful associations among entity types.
Relationship occurrence	Uniquely identifiable association, which includes one occurrence from each participating entity type.
Degree of a Relationship	Number of participating entities in relationship. It is limited by constraints such as cardinality and participation (mandatory {All}, optional {Not All}, disjoint {or}), nondisjoint {and})
Relationship of degree	A degree of relationship, for examples are : binary, ternary, quaternary or n-nary relationship

■ ERD BASIC CONCEPTS

CONCEPTS	DESCRIPTIONS
Attributes	A descriptive property or characteristics of an entity. Synonyms include element, property and field. In example as Student (Entity), StudentID (Attribute). There are many type of attributes such as simple, composite, single-valued, multi-valued and derived attribute.
Keys	An attribute, or a group of attributes, that assumes a unique value for each entity instance. It is sometimes called an identifier. There are candidate key, primary key, alternate key, foreign key and composite key.
Generalization	A concept wherein the attributes that are common to several types of an entity (inheritance relationship)
Supertype	An entity whose instances store attributes that are common to one or more entity subtypes.
Subtype	An entity whose instances may inherit common attributes from its entity supertype.

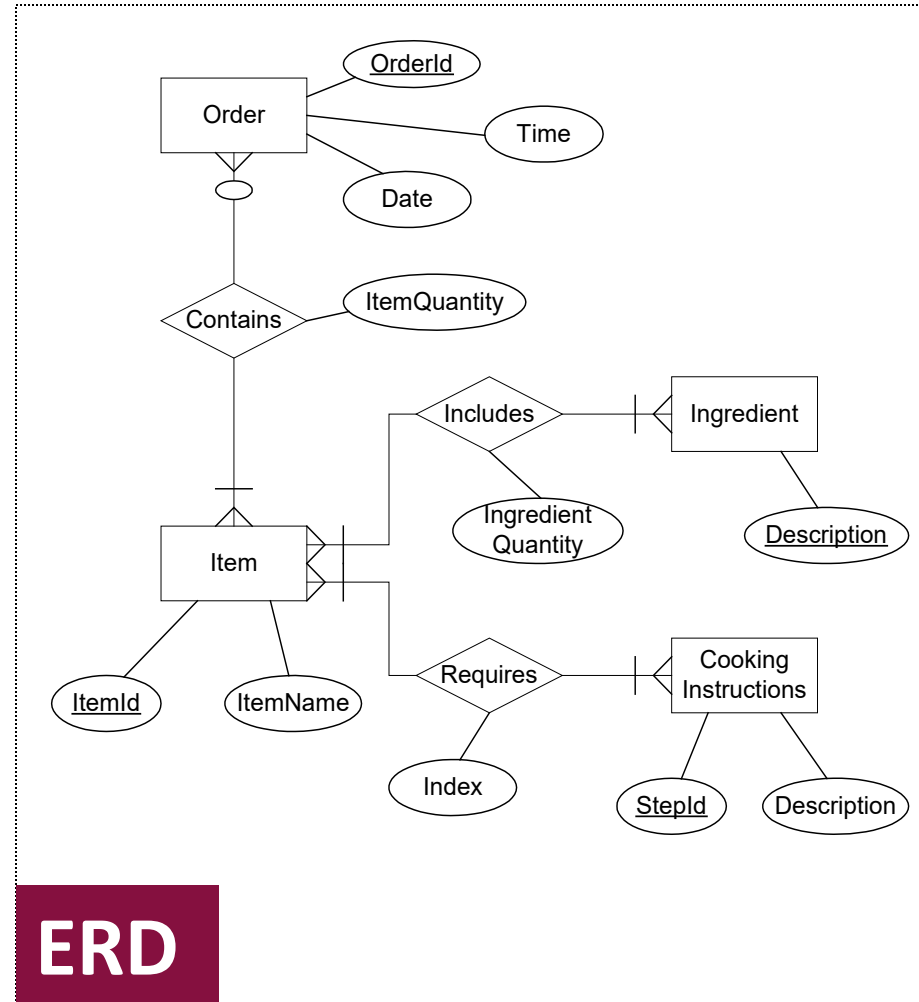
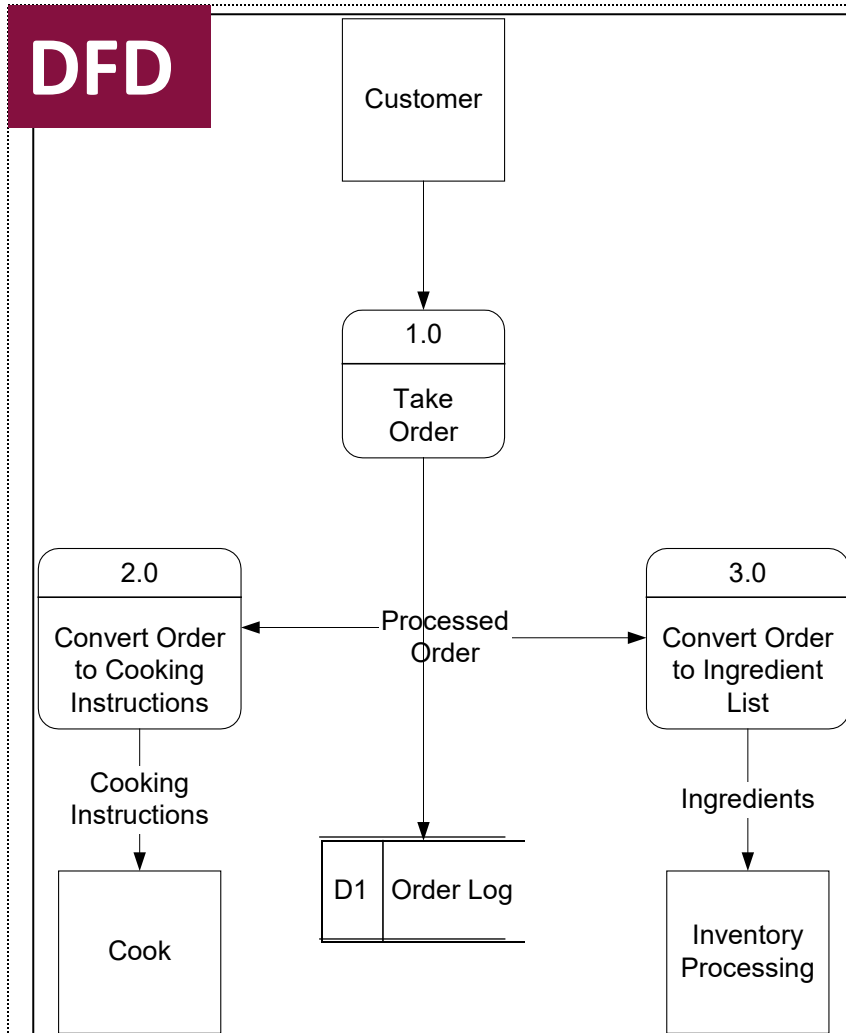
■ DFD & ERD

- DFDs and ERDs are both used to model systems, but they show two very different perspectives on the system.
- A DFD shows what the system *does* as well as the *data* that the system manipulates.
- An ERD shows **only** the *data* that the system manipulates.

■ DIFFERENCE BETWEEN DFD & ERD

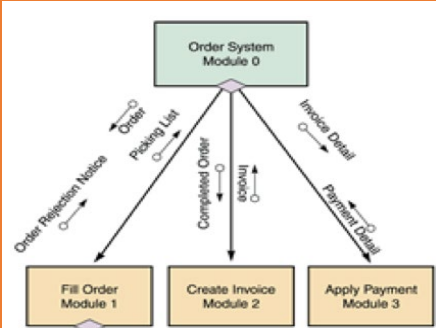
DFD	ERD
DFD shows how data enter a system, is transformed in that system, and how it is stored in it.	ERD represents the entity model and will show what a system or a database will look like but not explain how to implement it.
DFD shows what “things” or procedures transform inputted data into some output, as well as WHERE data is stored.	ERDs model WHAT kind of data and what relationships do different pieces of data have with each other. It would define the structure of our database tables.
DFDs main objective is to represent the processes and data flow between them.	ERDs main objective is to represent the data object or entity and the relationship between them.
DFD explains the flow and process of data input, data output, and storing data.	ERD explains and represents the relationship between entities stored in a database.
DFD is usually processed individually based on certain aspects, where each of them could be linked interactively.	ERD often represents a group of similar things; it shows how the data units are grouped into entities logically.
DFD is not enough to completely describe a system; it cannot also specify computations in a process.	ERD does not show the interaction between the model or data and how it changes in a system.

EXAMPLE OF DFD AND ERD



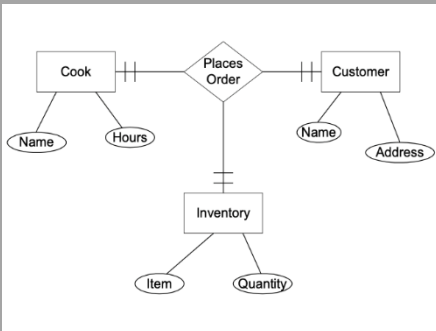
■ WHEN SHOULD BE USED?

STRUCTURE CHART



When we want to design a program code, usually we will **transform logical DFD** (to-be) to a set of structure chart. From a structure chart, we can write a pseudocode or draw a flow chart before to start programming.

ER DIAGRAM



We will need this very useful data modeling tool when to design database. **From DFD**, we have many data storage elements which will may to be automated. We use E-R diagram to design database for them. Meaning, how to systematically store all data as computer format.



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Thank You

update: August 2019 (sharinhh)

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